

The Closed Algal Loop

A Photobioreactor That Turns Exhaled Breath Back into Air and Food

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Abstract

A crew far from Earth cannot keep buying its air. Every breath consumes oxygen and produces carbon dioxide, and a mission that resupplies these from home is tethered to home; a mission that scrubs the carbon dioxide and throws it away wastes both the carbon and the effort. The living world solved this problem billions of years ago, and it solved it with photosynthesis: plants and algae take in carbon dioxide and light and give back oxygen, closing the loop that keeps a biosphere breathing. This study describes a machine that puts that ancient process to work for a spacecraft. The AL-1 'Chlorella' is a photobioreactor — a tank of illuminated microalgae — sized to consume a crew's exhaled carbon dioxide and regenerate their oxygen, while growing edible biomass as a by-product. It is regenerative life support: air that renews itself instead of air that runs out. The idea is not new, and that is its strength — closed ecological life support has been pursued for decades, from the Soviet BIOS-3 experiments that kept crews breathing on algae and plants to the European MELiSSA programme. We describe the biology, the reactor, the honest gap between a partial and a fully closed loop, and why algae are the natural first organism to fly.

1. Introduction: You Cannot Keep Buying Air

Life support on spacecraft has, until now, largely been a matter of stores and scrubbers. Oxygen is carried or generated by splitting water; exhaled carbon dioxide is captured by chemical scrubbers and, on the most advanced systems, partly recycled; and the balance is topped up from Earth. This works for missions measured in months and anchored by resupply, but it does not scale to the journeys and settlements this catalogue imagines. A crew a year from home, or a base meant to last decades, cannot depend on a supply line for the very air it breathes. It needs a life-support system that regenerates rather than consumes — one that turns the waste of breathing back into breathable air, indefinitely, on site.

Nature's version of this system is the biosphere, and its engine is photosynthesis. In the great planetary loop, animals consume oxygen and exhale carbon dioxide, and photosynthetic organisms consume that carbon dioxide and release oxygen, powered by light; the two halves balance, and the air stays breathable. A spacecraft is, in miniature, an incomplete biosphere: it has the animal half, the crew, but not the photosynthetic half. The Closed Algal Loop supplies the missing half in the most compact, fast-growing, and controllable form available — a reactor full of microalgae.

2. Concept Description

The AL-1 'Chlorella' is a photobioreactor: a transparent or illuminated vessel containing a dense suspension of microalgae in water, through which the cabin's air is circulated. The algae, bathed in light, take up the carbon dioxide the crew exhales and, by photosynthesis, release oxygen back into the air stream — the same gas exchange a forest performs, concentrated into a tank. As they do so the algae

grow, multiplying their mass, so a portion of the culture is continuously harvested to keep the population stable, and that harvested biomass is itself a resource: many microalgae are edible and nutritious, so the machine that renews the air also yields food.

The choice of algae over higher plants is deliberate. Microalgae grow far faster than crops, exchange gas more efficiently per unit volume, tolerate a compact and controllable reactor rather than requiring beds of soil and space, and can be managed as a fluid — pumped, dosed, and harvested like a chemical process rather than tended like a garden. For the specific job of closing the air loop in the least volume, algae are the strongest first candidate, which is why closed-life-support research has returned to them repeatedly. Higher plants, with their food and psychological benefits, come later and alongside; the algal loop is the compact core that keeps the air alive.

3. The Biology That Does the Work

Photosynthesis is the most consequential chemical process in the biosphere: using the energy of light, photosynthetic organisms combine carbon dioxide and water into sugars and release oxygen. Reversed, respiration by the crew consumes oxygen and sugars and releases carbon dioxide. Placed together in a closed volume with a light source, these two processes form a loop — the crew's exhaled carbon feeds the algae, the algae's released oxygen feeds the crew — and light, ultimately from the sun or from electric lamps, is the energy that drives the whole cycle. This is not a hypothesis but the operating principle of every ecosystem on Earth, scaled down to a tank.

The idea of harnessing it for life support has a long and serious pedigree. The Soviet BIOS-3 facility in the 1970s and 1980s sealed human crews inside a closed environment in which algae and later higher plants regenerated air and water and produced part of the food, demonstrating that a bioregenerative loop could sustain people for months (Salisbury, Gitelson and Lisovsky, 1997). The European Space Agency's MELiSSA project has pursued the same goal for decades as an engineered, multi-compartment micro-ecosystem, with photosynthetic organisms among its central elements, working steadily toward a loop closed enough to support a crew (Hendrickx et al., 2006). The Closed Algal Loop is not a departure from this lineage but a distillation of it: the photosynthetic compartment, built as a stand-alone air-renewing machine.

4. The Machine Itself

A photobioreactor is, in engineering terms, a vessel that keeps algae illuminated, fed, and mixed while gas is exchanged across the culture. It needs light delivered efficiently to a dense suspension — the central difficulty, since algae near the surface shade those deeper in, so the reactor geometry must spread the culture thin or pipe light into it. It needs a way to bring carbon-dioxide-laden cabin air into contact with the culture and carry the oxygen-enriched air back out. It needs circulation to keep the algae suspended and evenly exposed, nutrient dosing to keep them fed, and a harvest system to remove excess biomass. And it needs control, because a living culture is not a fixed device: its rate of gas exchange rises and falls with light, temperature, and the health of the population, and it must be kept in balance with the crew's steady demand.

In microgravity these tasks are harder, because gas and liquid do not separate as they do under gravity — bubbles do not rise, and cultures do not settle — so the reactor must manage the mixing and the gas-liquid interface actively rather than relying on buoyancy. This is a real and specific engineering

challenge for a space photobioreactor, distinct from its terrestrial cousins, and it is one of the reasons the technology has been slower to fly than its long research history might suggest. It is, however, an engineering problem in fluid handling, not a barrier of biology.

Function	Input	Output
Air renewal	cabin CO2 + light	O2 back to cabin
Biomass growth	CO2, water, nutrients	edible algae (food)
Water role	culture medium	recycled, retained
Energy source	sunlight or lamps	drives the whole loop
Harvest	excess algae	food + stable population

Table 1. The loop's flows — waste in, air and food out.

5. Operations Concept

Phase one is air-quality support: a photobioreactor that supplements existing life support, absorbing a share of the cabin's carbon dioxide and adding oxygen, reducing the load on chemical scrubbers and the quantity of consumables that must be carried — valuable immediately, and forgiving because the conventional system remains the backstop. Phase two grows the algal loop until it carries the primary burden of air regeneration for a crew, with the harvested biomass entering the food supply. Phase three integrates it with the higher-plant, waste-processing, and water-recovery compartments of a full bioregenerative system in the MELiSSA tradition (Hendrickx et al., 2006), approaching a genuinely closed loop in which air, water, and much of the food cycle continuously with minimal resupply. Each phase is useful alone, and the conventional system is never removed until its living replacement has earned the trust.

6. Honest Difficulties

A living life-support system trades the predictability of chemistry for the resilience — and the fragility — of biology. A culture can crash: infection, a toxin, a control failure, or a swing in temperature or light can sicken or kill the algae, and a life-support system that can die is one that must be buffered by reserves and, for a long time, by a conventional backup. Closing the loop fully is genuinely hard: BIOS-3 and MELiSSA between them represent decades of effort precisely because balancing every input and output of a self-contained ecosystem — trace nutrients, waste products, the exact match of gas production to consumption — is far more intricate than closing the oxygen-carbon exchange alone (Salisbury, Gitelson and Lisovsky, 1997; Hendrickx et al., 2006). The microgravity fluid problem is real. And the system demands continuous energy for light and continuous attention to a living population. None of this is a barrier of biology or physics — the loop runs on Earth every day — but a fully closed, reliable, space-rated version is a decades-long engineering programme, not a single device, and honesty requires saying so.

7. Pathway to Reality

The foundations are exceptionally solid. Photosynthesis needs no proof; microalgae are cultivated at industrial scale on Earth for food, feed, and chemicals; closed bioregenerative life support has been demonstrated with human crews in BIOS-3 and pursued rigorously for decades in MELiSSA (Salisbury,

Gitelson and Lisovsky, 1997; Hendrickx et al., 2006); and photobioreactors have already been flown in early experiments aboard orbiting stations to study algae in space. The credible near-term step is the supplementary air-quality reactor — a bounded, backstopped application that reduces consumables without betting the crew's air on the biology — with the loop progressively closed as reliability is proven. This concept grows directly from the harvester, greenhouse, and water-recovery infrastructure the rest of the catalogue assumes, and is among the most essential of all of them, because a settlement that cannot renew its own air is not a settlement at all.

8. Conclusion

Every breath a crew takes is a small withdrawal from a finite account, and a civilisation cannot live off-world by carrying the account with it. The Closed Algal Loop repays the account continuously, handing the crew's exhaled carbon to a tank of living green cells and taking back the oxygen those cells release, as the living world has done since before there were animals to breathe. It will not close perfectly at first, and the honest path keeps a conventional system alongside it for a long time; but every fraction of the air it renews is a fraction the settlement no longer has to buy. In the end the machines that let humanity live among the stars may be less like engines than like gardens — and the first of those gardens will be a tank of algae, quietly turning yesterday's breath back into tomorrow's.

References

Harvard referencing style (Cite Them Right).

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